

Discovering new classes at 2σ : the sigma-currency campaign

exps 146–151: CIFAR-10, CIFAR-100 (leakage-free) and Galaxy10 — summary and complete tables

P. C. Harris

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The measurement

Every number in this document is one currency: $f^*(2\sigma)$, the smallest fraction of a held-out class injected into a 5000-point corpus at which the *median expected significance* of a dataset-level two-sample test reaches 2σ against a 200-toy null (50 signal toys per fraction, add-one empirical tail, log-interpolated between grid fractions; lower is better). It is the discovery question asked the way an experiment would ask it: how rare can the new class be before this pipeline sees it at 2σ ? Tests: mean min-Euclidean distance, mean min-Mahalanobis (*Maha.*), 3-bandwidth MMD², the SparKer NP kernel ensemble, and — post-discovery only — two *anchor-aware* statistics that consume the anchors the discovery loop plants: the mean of $d_{\text{seen}} - d_{\text{disc}}$ (*Eucl. (anch.)*) and SparKer with kernels seeded at the discovered anchors (*SparKer (anch.)*).

Coverage

CIFAR-10, leakage-free: all 8 objectives of the design cube plus 16 residual constructions, trained from random initialisation (the encoder never saw the held-out class), 4 holdout draws; discovery mode under three loop variants (distance pool with the 95th-percentile cut; distance and density-ratio pools with the paper’s derived label-free cut, skip-on-refuse) and an $n_{\min}$ sweep. *Galaxy10*: 14 spaces (6 fine-tuned objectives, 2 transductive GCD arms, 6 residual constructions) on three pretrained backbones, 5 holdout draws each, pre-discovery; discovery mode (density-ratio pool, derived cut, $n_{\min}=5$) on the three supervised arms and the three cross-draw residual children. *CIFAR-100* (same from-scratch lineage, $b=0.01$): all 8 objectives plus 16 residual constructions, 4 draws, pre-discovery. Every cell is single-seed; the campaign’s variance decomposition (seed spread an eighth of draw spread) is why intervals are over draws.

Findings

1. **Neither the probe nor ranking AUC predicts f^* .** The best-probed spaces are rarely the most detectable (CIFAR: probe 0.97 spaces sit mid-table; NPLM-dist. probes at 0.79 yet detects at 0.027), so representation quality for discovery has to be measured in this currency, not proxied.
2. **The best anchor-blind test is dataset-dependent, but always a density-ratio test.** SparKer leads on the compact CIFAR scratch heads (best or tied on 17/20 spaces, never censored); MMD leads on the Galaxy10 transfer heads (essentially never censored there, while SparKer and Mahalanobis are censored on 1–4 of 5 draws for half the rows).
3. **The residual construction earns a place, and its role scales with class count and with discovery.** Frozen, it is a stabiliser at 10 classes, and at 100 classes the concat sets the record (SupCon→res 0.018 vs. parent 0.023). Through the discovery loop, the *contrastive* child of SupCon becomes the best pipeline on 2 of 3 Galaxy10 backbones (0.013/0.015 vs. parents’ 0.018/0.020) and rescues the one CIFAR cell where the parent’s critic abstains entirely

(holdout 4: child engages at purity 0.2–0.4 and reaches 0.015). The NPLM child ties its parent throughout: it is the contrastive residual that recovers discarded novelty.

4. **Discovery pays only through the anchors it plants.** Anchor-blind tests *degrade* after the fine-tuning loop on both datasets; the anchor-aware statistics convert the same loop into the best pipelines measured. The loop beats the frozen space in 38/48 CIFAR (arm, draw, variant) cells and in all 9 Galaxy10 (backbone, arm) blocks.
5. **The pool decides, and the density-ratio pool with the derived cut supplies it.** Where the distance pool gives SupCon purity 0.000 (and pathological anchors), the np pool gives 0.3–0.9 on CIFAR and median purity 1.00 on Galaxy10 — and f^* follows. The scorer, not the cut, is what makes the pool pure; the cut sets its size and, when the estimated novelty is below $n_{\min}$, *declines* — every remaining loss is such an abstention (daggers in the tables), and the one arm that declines everywhere (SupCon at holdout 4, all $n_{\min}$) is exactly the arm whose pools are irreparably impure. The $n_{\min}$ sweep says: engage as small as the clustering bears ($n_{\min}=5$), at no purity cost.
6. **The chosen algorithm** (Table 3): a supervised contrastive space (or, best, its contrastive residual child), the density-ratio pool with the derived label-free cut, and the anchor-aware Euclidean statistic. It wins on every dataset/backbone ($1.3\text{--}1.7\times$ over the best frozen pipeline), and on Galaxy10/LeJEPa the residual-child pipeline reaches $f^*=0.013$ — *equal to* the transductive GCD ceiling, without ever seeing an unlabelled novel image.
7. **Diffuse shift is the measured boundary of interpretability** (Table 32). A systematic blur of seen-class images is almost perfectly decodable (probe 0.86–1.00) yet the fine-tuned discovery spaces barely register it: the density-ratio tests censor on most cells and mean-shift Mahalanobis — the loser on class novelty — becomes the only reliable detector, while the raw pretrained trunk sees the smear at $f^*=0.006\text{--}0.010$ everywhere. SparKer prefers the one-label (semi-clustered) smear in every engaged comparison: the kernel tests’ sharpness for class novelty is a dependence on clusteredness, and a shift with no class structure is detectable but yields nothing an anchor could mean. Practical corollary: pair the discovery pipeline with a mean-shift monitor on the frozen trunk. The on-manifold corner (Table 34) is the mirror image: fine-grained novelty (aircraft variants, car models) embeds *inside* the seen manifold — distance ranking falls below chance, per-event power is dead, pools starve — yet the class is still a dense clump, so SparKer’s local kernels detect it ($f^*=0.012\text{--}0.021$ on cars) while Mahalanobis censors. Clustered-off-manifold: anchors interpret. Clustered-on-manifold: locality detects, nothing anchors. Diffuse: mean-shift detects, nothing anchors.
8. **The holdout draw dominates the uncertainty** everywhere (ranges span $3\text{--}5\times$ within a row; the loop amplifies draw variance $\sim 5\times$), so single-draw numbers are never quoted without their spread, and per-draw points are tabulated in full.

	clustered novelty	diffuse shift
off-manifold	everything works; anchors interpret. Best pipelines $f^*=0.008$ – 0.020 (Tables 3, 27)	detected only by the mean-shift monitor on the <i>frozen trunk</i> ($f^*=0.006$ – 0.010); head spaces nearly blind (Tables 32, 33); yet the <i>discovery loop finds it</i> : the np pool isolates the smeared events in every cell (purity often 1.00), the fine-tune pulls them off the manifold, and anchor-aware tests reach $f^*\approx 0.01$ – 0.02 (Table 36)
on-manifold	SparKer’s local kernels detect the clump ($f^*=0.012$ – 0.039) but distance ranking is below chance, pools starve, Mahalanobis censors: detection without anything to anchor (Table 34). Fine-tuned heads + multi-holdout soften it: aircraft engages at $f=0.05$ (purity 0.54–1.00); cars engages only through the res-nplm child, and discovery never beats frozen locality (Table 35)	the doubly-hard corner, covered by both failure modes

Table 1: Where discovery works, where only detection works, and where the method abstains: the failure 2×2 , every cell measured in the $f^*(2\sigma)$ currency. Locality (SparKer) detects clumps; mean-shift (Mahalanobis) detects smears; anchors interpret only what is both clustered and outlying — and the pipeline reports when it cannot. *Coverage*: Cells summarise Tables 3–34; the derived label-free cut abstains wherever estimated novel mass is below clusterability, which governs the boundary between cells.

The campaign in one table

arm	draw	frozen SparKer	dist/95th	dist/derived	np/derived
SupCon	h4	0.015	0.027 (MMD)	0.014 (MMD) [†]	0.014 (MMD) [†]
	h7	0.021	0.035 (SparKer (anch.))	0.033 (SparKer (anch.))	0.018 (SparKer)
	h8	0.018	0.016 (Eucl. (anch.))	0.017 (MMD)	0.011 (Eucl. (anch.))
	h9	0.018	0.008 (Eucl.)	0.011 (Eucl. (anch.))	0.009 (SparKer)
SupCon+SIGReg ($\lambda=5$)	h4	0.042	0.021 (Eucl. (anch.))	0.028 (MMD)	0.028 (Eucl.)
	h7	0.034	0.021 (Eucl. (anch.))	0.036 (SparKer) [†]	0.036 (SparKer) [†]
	h8	0.040	0.015 (MMD)	0.021 (MMD)	0.020 (Eucl. (anch.))
	h9	0.032	0.016 (Eucl. (anch.))	0.026 (MMD) [†]	0.026 (MMD) [†]
NPLM-dist. (sup.)	h4	0.040	0.024 (Eucl. (anch.))	0.037 (Eucl. (anch.))	0.033 (Eucl. (anch.))
	h7	0.029	0.022 (Eucl. (anch.))	0.039 (Eucl. (anch.))	0.021 (Eucl. (anch.))
	h8	0.024	0.016 (SparKer (anch.))	0.015 (Eucl. (anch.))	0.016 (Eucl. (anch.))
	h9	0.024	0.009 (SparKer (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
NPLM-dist.+classwise SIGReg	h4	0.025	0.019 (Eucl. (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
	h7	0.029	0.018 (Eucl. (anch.))	0.027 (SparKer (anch.))	0.011 (Eucl. (anch.))
	h8	0.024	0.015 (Eucl. (anch.))	0.012 (Eucl. (anch.))	0.009 (Eucl. (anch.))
	h9	0.028	0.016 (Maha.)	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))

Table 2: Does the discovery loop beat the frozen space? Frozen-space SparKer $f^*(2\sigma)$ against the best post-discovery test (named in parentheses) under three loop variants: the campaign’s distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper’s construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; † = the derived cut had declined at the crossing fraction, so that number is the frozen space’s and is not counted as a win; ‘–’ = variant not run or still running.

Summary tables

dataset / backbone	best frozen (space / test, median f^*)		best discovery (arm / test)		gain
CIFAR-100 (scratch, 4 draws)	SupCon→res (concat) / SparKer	0.018	SupCon→res-nplm / SparKer ^e	0.008	2.2×
CIFAR-10 (scratch, 4 draws)	SupCon→res-nplm (resid.) / SparKer	0.017	SupCon / Eucl. (anch.)	0.013	1.3×
Galaxy10 / lejepa	SupCon / MMD	0.023	SupCon→res / Eucl. (anch.)	0.013	1.7×
Galaxy10 / visreg	SupCon→res (concat) / MMD	0.026	SupCon→res / Eucl. (anch.)	0.015	1.7×
Galaxy10 / dino	SupCon+SIGReg / MMD	0.029	SupCon / Eucl. (anch.)	0.019	1.5×
transductive ref. (lejepa)	gcd-ft / Maha.	0.013	–	–	sees the class
transductive ref. (visreg)	gcd-ft / Maha.	0.015	–	–	sees the class
transductive ref. (dino)	gcd-ft / MMD	0.028	–	–	sees the class

Table 3: The chosen algorithm, per dataset: best frozen space vs. best discovery pipeline in the same currency. Discovery wins everywhere, always through an anchor-aware test; the transductive GCD rows are the ceiling given the whole novel class unlabelled. *Coverage:* Medians over draws (censored cells imputed at 0.15); CIFAR-10 discovery = np pool + derived cut ($n_{\min}=10$), Galaxy10 discovery = np pool + derived cut ($n_{\min}=5$), both skip-on-refuse; frozen candidates exclude the transductive GCD arms (shown separately). Declined-cut crossings are excluded from the discovery medians; ^e = CIFAR-100 medians cover the engaged draws only (the cut abstains on 33/48 cells at $b=0.01$).

The discovery cut: tight vs. $s/\sqrt{b}$ vs. $\max(\text{tv}, \text{BBE})$

The settled pipeline’s pool cut is the label-free *tight* cut: it keeps the tightest threshold still holding $n_{\min}$ estimated-novel points, and *declines* below that. Exp 138 showed its base-rate estimate $\hat{b}$ (a TV-sum of density-ratio weights) is biased low on overlapping / low-rate spaces, so the cut over-abstains. Two remedies were measured in the same $f^*(2\sigma)$ currency: the significance-optimal Punzi cut $s/\sqrt{b}$ (which never abstains), and feeding the tight cut a less-biased $\hat{b} = \max(\text{tv}, \text{BBE})$ (BBE = best-bin mixture-proportion estimation, Garg et al. 2021), which only loosens the decline when a second, independent estimator confirms novelty.

The verdict (Table 4, per-arm detail Tables 5–7):

- **The tight cut is tightest where it fires but abstains often** — on CIFAR-100 it engages on only 9/48 cells (declining every residual child on every draw), and even on the clean datasets it leaves the weak arms (nplm-sup-ft, the residual children) empty. Where it does engage its per-cell f^* is best (CIFAR-100 median 0.011), but that covers a minority of cases.
- **$s/\sqrt{b}$ and $\max(\text{tv}, \text{BBE})$ are statistically interchangeable** — near-identical engagement, anchor-aware rate, best and median f^* on every dataset. BBE was therefore a valuable *diagnostic* (it proved the tight cut’s abstention is a $\hat{b}$ -bias artifact, not a true “nothing here”), but its extra machinery buys nothing over the simpler significance-optimal cut.
- **$s/\sqrt{b}$ wins-or-ties on median everywhere** and carries the most anchor-aware (interpretable) discoveries. On the clean-pool datasets (CIFAR-10, Galaxy) essentially every detection is anchor-aware regardless of cut; only CIFAR-100’s $b=0.01$ regime splits into $\sim$ half anchor-aware, half blind-detection.

Recommendation: **$s/\sqrt{b}$ is the default cut** — simplest (one rule, no second estimator), never abstains, and equal-or-better than both alternatives on the achievable f^* ; the tight cut is retained only when a guaranteed-pure, self-abstaining pool is specifically wanted.

dataset	cut	engage	anchor-aware	best f^*	median f^*
<i>CIFAR-10</i>	tight	50/52	27/52	0.009	0.020
	$s/\sqrt{b}$	52/52	42/52	0.008	0.017
	max(tv,BBE)	52/52	40/52	0.007	0.018
<i>CIFAR-100</i>	tight	9/48	6/48	0.008	0.011
	$s/\sqrt{b}$	48/48	25/48	0.009	0.019
	max(tv,BBE)	47/48	24/48	0.009	0.019
<i>Galaxy/lejepa</i>	tight	32/35	30/35	0.010	0.020
	$s/\sqrt{b}$	35/35	33/35	0.010	0.015
	max(tv,BBE)	35/35	33/35	0.010	0.015
<i>Galaxy/visreg</i>	tight	34/35	28/35	0.010	0.020
	$s/\sqrt{b}$	35/35	33/35	0.010	0.016
	max(tv,BBE)	35/35	33/35	0.010	0.018
<i>Galaxy/dino</i>	tight	29/35	27/35	0.014	0.020
	$s/\sqrt{b}$	35/35	32/35	0.012	0.019
	max(tv,BBE)	35/35	35/35	0.013	0.021

Table 4: The discovery-cut comparison. The tight cut is tightest *where it fires* but abstains often (esp. on the residual children and the low-rate CIFAR-100 regime); $s/\sqrt{b}$ and max(tv,BBE) both engage nearly everywhere and are statistically interchangeable, with $s/\sqrt{b}$ marginally ahead (more anchor-aware wins, equal or lower median f^*) while being the simpler rule. On the clean-pool datasets (CIFAR-10, Galaxy) essentially every detection is anchor-aware; only CIFAR-100’s $b=0.01$ regime splits into $\sim$ half anchor-aware, half blind. *Coverage:* 15 (dataset, cut) rows; each aggregates all (arm, draw) cells (CIFAR: 16/12 arms $\times$ 4 draws; Galaxy: 7 arms $\times$ 5 draws). *engage* = cells with a finite f^* ; *anchor-aware* = of those, how many are won by an anchor-using test (eucl-disc / sparker-anch) rather than a blind two-sample test; best/median over engaged cells. np density-ratio pool, $n_{\min}=5$ throughout.

method	tight	$s/\sqrt{b}$	max(tv,BBE)
SupCon	0.011 [0.009–0.018]	0.015 [0.008–0.016]	0.014 [0.007–0.023]
SupCon+SIGReg	0.036 [0.020–0.072]	0.020 [0.014–0.025]	0.022 [0.016–0.025]
NPLM-dist.	0.021 [0.016–0.033]	0.019 [0.016–0.025]	0.022 [0.016–0.025]
NPLM+cw-SIGReg	0.015 [0.009–0.020]	0.014 [0.009–0.021]	0.015 [0.012–0.016]
SupCon $\rightarrow$ res	0.015 [0.009–0.016]	0.014 [0.006–0.016]	0.015 [0.008–0.019]
SupCon $\rightarrow$ res-nplm	0.016 [0.009–0.021]	0.016 [0.006–0.021]	0.014 [0.012–0.021]
SupCon+SIGReg $\rightarrow$ res	0.029 [0.027–0.041]	0.015 [0.014–0.024]	0.018 [0.017–0.021]
SupCon+SIGReg $\rightarrow$ res-nplm	0.017 [0.010–0.020]	0.016 [0.014–0.024]	0.016 [0.013–0.021]
NPLM-dist. $\rightarrow$ res	0.024 [0.020–0.031]	0.016 [0.014–0.017]	0.018 [0.015–0.022]
NPLM-dist. $\rightarrow$ res-nplm	0.033 [0.016–0.072]	0.018 [0.015–0.037]	0.022 [0.021–0.038]
NPLM+cw-SIGReg $\rightarrow$ res	0.030 [0.011–0.044]	0.016 [0.009–0.017]	0.015 [0.014–0.017]
NPLM+cw-SIGReg $\rightarrow$ res-nplm	0.034 [0.020–0.048]	0.015 [0.014–0.021]	0.016 [0.014–0.024]
SupCon+cw-SIGReg	0.026 [0.020–0.036]	0.022 [0.018–0.032]	0.019 [0.017–0.023]
SupCon+SIGReg (rep.)	0.037 [0.029–0.100]	0.025 [0.020–0.028]	0.024 [0.021–0.026]
SupCon+SIGReg $\rightarrow$ res	0.019 [0.017–0.020]	0.019 [0.017–0.024]	0.019 [0.017–0.024]
SupCon+SIGReg $\rightarrow$ res-nplm	0.023 [0.015–0.025]	0.022 [0.015–0.025]	0.022 [0.019–0.023]

Table 5: CIFAR-10: per-method f^* under the three cuts. The tight cut’s $>.1$ rows are arms it declines on every draw — overwhelmingly the residual children — which $s/\sqrt{b}$ and max(tv,BBE) recover. *Coverage:* CIFAR-10: per-method best- f^* median [min–max] over 4 draws, three cuts; ‘ $>.1$ ’ = the method never reached 2σ by $f=0.1$ under that cut (the tight cut’s blanks are its abstentions).

method	tight	$s/\sqrt{b}$	$\max(\text{tv}, \text{BBE})$
SupCon	0.018 [0.018–0.018]	0.019 [0.011–0.024]	0.018 [0.012–0.025]
SupCon+SIGReg	0.011 [0.011–0.011]	0.019 [0.015–0.024]	0.015 [0.015–0.025]
NPLM-dist.	0.014 [0.010–0.018]	0.018 [0.015–0.029]	0.019 [0.016–0.031]
NPLM+cw-SIGReg	0.010 [0.008–0.025]	0.021 [0.013–0.027]	0.022 [0.009–0.028]
SupCon $\rightarrow$ res	>.1	0.021 [0.012–0.030]	0.020 [0.011–0.025]
SupCon $\rightarrow$ res-nplm	0.008 [0.008–0.008]	0.021 [0.009–0.025]	0.021 [0.009–0.025]
SupCon+SIGReg $\rightarrow$ res	>.1	0.018 [0.015–0.025]	0.019 [0.014–0.027]
SupCon+SIGReg $\rightarrow$ res-nplm	0.014 [0.014–0.014]	0.016 [0.013–0.027]	0.021 [0.012–0.027]
NPLM-dist. $\rightarrow$ res	>.1	0.020 [0.010–0.026]	0.019 [0.013–0.029]
NPLM-dist. $\rightarrow$ res-nplm	>.1	0.021 [0.013–0.026]	0.017 [0.014–0.027]
NPLM+cw-SIGReg $\rightarrow$ res	>.1	0.021 [0.016–0.034]	0.023 [0.016–0.026]
NPLM+cw-SIGReg $\rightarrow$ res-nplm	>.1	0.022 [0.017–0.033]	0.017 [0.016–0.027]

Table 6: CIFAR-100: per-method f^* under the three cuts. The tight cut’s >.1 rows are arms it declines on every draw — overwhelmingly the residual children — which $s/\sqrt{b}$ and $\max(\text{tv}, \text{BBE})$ recover. *Coverage:* CIFAR-100: per-method best- f^* median [min–max] over 4 draws, three cuts; ‘>.1’ = the method never reached 2σ by $f=0.1$ under that cut (the tight cut’s blanks are its abstentions).

method	tight	$s/\sqrt{b}$	$\max(\text{tv}, \text{BBE})$
SupCon	0.018 [0.010–0.020]	0.012 [0.010–0.015]	0.014 [0.010–0.021]
SupCon+SIGReg	0.016 [0.010–0.020]	0.015 [0.010–0.022]	0.014 [0.010–0.021]
NPLM-dist.	0.037 [0.020–0.100]	0.015 [0.010–0.027]	0.016 [0.010–0.034]
SupCon $\rightarrow$ res	0.013 [0.010–0.023]	0.014 [0.010–0.023]	0.015 [0.012–0.024]
SupCon $\rightarrow$ res-nplm	0.020 [0.010–0.036]	0.014 [0.010–0.025]	0.014 [0.012–0.036]
SupCon+SIGReg $\rightarrow$ res	0.018 [0.011–0.025]	0.018 [0.012–0.022]	0.016 [0.010–0.019]
SupCon+cw-SIGReg	0.030 [0.011–0.040]	0.021 [0.012–0.035]	0.021 [0.011–0.028]

Table 7: Galaxy/LeJEPa: per-method f^* under the three cuts. The tight cut’s >.1 rows are arms it declines on every draw — overwhelmingly the residual children — which $s/\sqrt{b}$ and $\max(\text{tv}, \text{BBE})$ recover. *Coverage:* Galaxy/LeJEPa: per-method best- f^* median [min–max] over 5 draws, three cuts; ‘>.1’ = the method never reached 2σ by $f=0.1$ under that cut (the tight cut’s blanks are its abstentions).

CIFAR-10: pre-discovery

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg ($\lambda=5$)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047 ¹	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034 ¹	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046
SupCon+SIGReg (repulse)	0.901	0.10	0.074 ¹	0.041	0.039	0.026–0.043
→ res (residual)	0.921	0.02	0.062 ¹	0.044	0.039	0.033–0.064
→ res (concat)	0.940	0.11	0.037	0.035	0.038	0.031–0.040
→ res-nplm (residual)	0.900	0.11	0.053	0.043	0.043	0.024–0.049
→ res-nplm (concat)	0.930	0.08	0.041	0.037	0.040	0.029–0.042
SimCLR	0.808	0.00	$> 0.1^1$	0.027	0.056	0.056–0.056
VISReg/LeJEPA	0.832	0.01	$> 0.1^1$	0.047	0.091	0.091–0.091
NPLM (unsup.)	0.799	0.05	$> 0.1^1$	0.086	$> 0.1^1$	–
SupCon+classwise SIGReg ($\lambda=5$)	0.895	0.13	0.060 ¹	0.033	0.058	0.039–0.070

Table 8: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 26/29 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.955	0.03	$> 0.1^2$	0.024	0.023	0.010–0.036
→ res (residual)	0.925	0.05	$> 0.1^3$	0.025	0.028	0.023–0.030
→ res (concat)	0.963	0.05	0.057 ¹	0.024	0.018	0.017–0.026
→ res-nplm (residual)	0.929	0.10	$> 0.1^3$	0.025	0.027	0.016–0.048
→ res-nplm (concat)	0.963	0.07	$> 0.1^3$	0.023	0.022	0.010–0.030
SupCon+SIGReg ($\lambda=5$)	0.947	0.05	0.055	0.022	0.023	0.014–0.026
→ res (residual)	0.906	0.07	$> 0.1^3$	0.028	0.027	0.025–0.032
→ res (concat)	0.959	0.05	0.056	0.020	0.019	0.016–0.026
→ res-nplm (residual)	0.932	0.05	0.055 ¹	0.022	0.024	0.016–0.027
→ res-nplm (concat)	0.941	0.05	0.059	0.021	0.023	0.014–0.026
NPLM-dist. (sup.)	0.857	0.07	$> 0.1^2$	0.025	0.034	0.017–0.046
→ res (residual)	0.892	0.11	$> 0.1^2$	0.026	0.026	0.025–0.033
→ res (concat)	0.917	0.10	$> 0.1^3$	0.026	0.028	0.026–0.041
→ res-nplm (residual)	0.905	0.08	$> 0.1^2$	0.032	0.026	0.021–0.031
→ res-nplm (concat)	0.924	0.07	$> 0.1^2$	0.028	0.025	0.022–0.029
NPLM-dist.+classwise SIGReg	0.944	0.12	$> 0.1^2$	0.021	0.025	0.015–0.029
→ res (residual)	0.911	0.06	$> 0.1^3$	0.025	0.027	0.024–0.030
→ res (concat)	0.956	0.06	$> 0.1^2$	0.025	0.027	0.023–0.032
→ res-nplm (residual)	0.878	0.04	$> 0.1^3$	0.027	0.027	0.026–0.030
→ res-nplm (concat)	0.955	0.04	$> 0.1^2$	0.022	0.026	0.018–0.030
SimCLR	0.799	0.01	$> 0.1^1$	0.046	0.052	0.052–0.052
VISReg/LeJEPa	0.797	0.03	$> 0.1^1$	0.059	0.047	0.047–0.047
NPLM (unsup.)	0.757	0.01	$> 0.1^1$	0.040	0.031	0.031–0.031
SupCon+SIGReg (repulse)	0.823	0.00	$> 0.1^1$	0.031	0.029	0.029–0.029

Table 9: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-100, leakage-free lineage, single holdout; draws 4, 43, 48, 57; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	> 0.1	0.072	0.027	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	0.026
→ res-nplm (residual)	0.016	0.08	0.082	0.029	0.065	0.045	0.037	0.031
SupCon+SIGReg ($\lambda=5$)	0.042	0.01	0.033	0.021	> 0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	0.035	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	0.017	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	> 0.1	0.024	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	> 0.1	> 0.1	0.081	0.035	0.043	0.041
→ res-nplm (residual)	0.049	0.01	> 0.1	0.047	0.087	0.062	0.083	0.044
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	0.019	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	0.027	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	0.022	0.037	0.036	0.047	0.041

Table 10: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

CIFAR-10: discovery mode, per draw and loop variant

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.03	0.042	0.039	0.068	0.039	0.040	0.032
→ res-nplm (residual)	0.016	0.01	0.047	0.039	0.049	0.034	0.027	0.024
SupCon+SIGReg ($\lambda=5$)	0.042	0.00	0.038	0.062	> 0.1	0.028	0.047	0.044
→ res (residual)	0.091	0.00	> 0.1	> 0.1	0.078	0.038	0.049	0.087
→ res-nplm (residual)	0.023	0.07	0.072	0.024	> 0.1	0.039	0.040	0.026
NPLM-dist. (sup.)	0.040	–	0.077	0.037	0.072	0.045	0.043	0.043
→ res (residual)	0.077	0.02	> 0.1	> 0.1	> 0.1	0.038	0.052	0.060
→ res-nplm (residual)	0.049	–	0.089	0.065	0.089	0.080	0.049 [†]	0.074
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020	0.074	0.029 [†]	0.028 [†]	0.021
→ res (residual)	0.084	–	> 0.1	0.050	0.083	0.041	0.047	0.050
→ res-nplm (residual)	0.079	–	0.049	0.050	0.046	0.043	0.067	0.050

Table 11: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.21	0.033	0.015	0.024	0.041	0.033	0.029
→ res-nplm (residual)	0.016	0.14	0.035	0.020	0.039	0.041	0.037	0.034
SupCon+SIGReg ($\lambda=5$)	0.042	0.13	0.028	0.030	0.083	0.034	0.042	0.030
→ res (residual)	0.091	0.15	0.082	0.030	0.033	0.042	0.060	0.047
→ res-nplm (residual)	0.023	0.12	0.044	0.018	0.074	0.037	0.041	0.021
NPLM-dist. (sup.)	0.040	–	0.087	0.033	0.084	0.049	0.073	0.042
→ res (residual)	0.077	0.43	0.056	0.020	0.029	0.046	0.046	0.034
→ res-nplm (residual)	0.049	–	0.089	0.046	0.089	0.080	0.049 [†]	0.062
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020	0.072	0.029 [†]	0.027 [†]	0.036
→ res (residual)	0.084	–	0.079	0.050	0.046	0.044	0.045	0.050
→ res-nplm (residual)	0.079	–	0.044	0.050	0.046	0.043	0.057	0.050
SupCon+classwise SIGReg ($\lambda=5$)	0.049	0.10	0.040	0.021	0.049	0.044	0.048	0.020
SupCon+SIGReg (repulse)	0.026	–	0.029 [†]	–	0.048 [†]	0.042 [†]	0.028 [†]	–
→ res (residual)	0.064	0.22	0.033	0.020	0.025	0.049	0.046	0.041
→ res-nplm (residual)	0.042	0.14	0.028	0.022	0.027	0.046	0.029	0.039

Table 12: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 16 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	> 0.1	> 0.1	0.043	0.047	0.052	0.035
SupCon+SIGReg ($\lambda=5$)	0.034	0.03	0.042	0.021	> 0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	0.022	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	0.018	0.024	0.037	0.028	0.025

Table 13: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071	> 0.1	0.036	0.056	0.040	0.033
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.079	0.036 †	–
NPLM-dist. (sup.)	0.029	0.05	> 0.1	0.039	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	0.027

Table 14: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022	0.020	0.021	0.052	0.018	0.020
→ res (residual)	0.026	0.63	0.017	0.014	0.016	0.049	0.018	0.017
→ res-nplm (residual)	0.019	0.61	0.021	0.021	0.021	0.065	0.022	0.023
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.072	0.036 [†]	–
→ res (residual)	0.089	0.12	0.059	0.030	0.027	0.079	0.066	0.046
→ res-nplm (residual)	0.025	0.63	0.039	0.016	0.038	0.044	0.028	0.018
NPLM-dist. (sup.)	0.029	0.18	0.044	0.021	0.033	0.043	0.041	0.029
→ res (residual)	0.045	–	0.044	0.050	0.031	0.076	0.047	0.050
→ res-nplm (residual)	0.049	–	0.092	–	0.089	0.073	0.072	–
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	0.011	0.018	0.049	0.033	0.019
→ res (residual)	0.039	0.22	0.036	0.030	0.023	0.068	0.029	0.030
→ res-nplm (residual)	0.052	–	0.068	0.050	0.026 [†]	0.072	0.048	0.050
SupCon+classwise SIGReg ($\lambda=5$)	0.067	–	0.031 [†]	–	0.056 [†]	0.071 [†]	0.054 [†]	–
SupCon+SIGReg (repulse)	0.035	–	0.037	0.050	0.043	0.076	0.034	0.088
→ res (residual)	0.043	0.33	0.027	0.020	0.019	0.084	0.036	0.027
→ res-nplm (residual)	0.043	0.07	0.029	0.025	0.028	0.056	0.030	0.028

Table 15: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 7, density-ratio pool, derived cut; 16 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	0.016	0.023	0.018	0.021	0.017
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.026	0.019	> 0.1	0.015	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	0.016
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	0.015	0.022	0.022	0.017	0.015

Table 16: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052	> 0.1	0.038	0.017	0.031	0.027
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.024	0.076	0.062	0.021	0.029	0.029
NPLM-dist. (sup.)	0.024	0.59	0.063	0.015	0.025	0.023	0.023	0.017
NPLM-dist.+classwise SIGReg	0.024	0.55	0.042	0.012	0.039	0.036	0.035	0.021

Table 17: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014	0.011	0.015	0.019	0.017	0.014
→ res (residual)	0.018	0.42	0.017	0.020	0.016	0.019	0.016	0.020
→ res-nplm (residual)	0.018	0.56	0.021	0.015	0.012	0.017	0.016	0.015
SupCon+SIGReg ($\lambda=5$)	0.040	0.22	0.024	0.020	0.024	0.025	0.032	0.035
→ res (residual)	0.044	–	0.046	0.050	0.045	0.021 [†]	0.041	0.050
→ res-nplm (residual)	0.021	0.48	0.038	0.020	0.025	0.021	0.024	0.036
NPLM-dist. (sup.)	0.024	0.64	0.046	0.016	0.041	0.024	0.018	0.017
→ res (residual)	0.046	0.30	0.028	0.030	0.028	0.022	0.028	0.030
→ res-nplm (residual)	0.029	0.18	0.027	0.020	0.026	0.027	0.027	0.028
NPLM-dist.+classwise SIGReg	0.024	0.73	0.075	0.009	0.037	0.044	0.040	0.021
→ res (residual)	0.054	0.40	0.026	0.011	0.019	0.019	0.025	0.024
→ res-nplm (residual)	0.033	0.36	0.074	0.020	0.027	0.022	0.037	0.020
SupCon+classwise SIGReg ($\lambda=5$)	0.070	0.05	0.071	0.030	0.059	0.026	0.066	0.064
SupCon+SIGReg (repulse)	0.043	0.17	0.029	0.030	0.030	0.017 [†]	0.037	0.065
→ res (residual)	0.034	0.31	0.025	0.020	0.020	0.017	0.022	0.023
→ res-nplm (residual)	0.049	0.19	0.034	0.030	0.025	0.018 [†]	0.028	0.030

Table 18: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 8, density-ratio pool, derived cut; 16 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	0.008	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg ($\lambda=5$)	0.032	0.01	0.025	0.016	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	0.009
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	0.016	0.029	0.017	0.017

Table 19: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.013	0.011	0.025	0.020	0.014	0.018
SupCon+SIGReg ($\lambda=5$)	0.032	−	0.047	> 0.1	0.048	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.82	0.036	0.020	0.024	0.029	0.021	0.027
NPLM-dist.+classwise SIGReg	0.028	0.51	0.026	0.020	0.028	0.036	0.026	0.020

Table 20: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.28	0.010	0.013	0.016	0.025	0.009	0.010
→ res (residual)	0.012	0.50	0.009	0.010	0.009	0.019	0.009	0.010
→ res-nplm (residual)	0.015	0.75	0.014	0.013	0.014	0.016	0.009	0.010
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	0.050	0.045	0.026 [†]	0.027 [†]	0.050
→ res (residual)	0.049	0.15	0.035	0.030	0.027	0.037	0.028	0.030
→ res-nplm (residual)	0.017	0.59	0.024	0.010	0.018	0.028	0.017	0.016
NPLM-dist. (sup.)	0.024	0.84	0.027	0.020	0.023	0.029	0.023	0.020
→ res (residual)	0.044	0.16	0.028	0.030	0.027	0.034	0.028	0.030
→ res-nplm (residual)	0.024	0.55	0.082	0.016	0.027	0.027	0.025	0.026
NPLM-dist.+classwise SIGReg	0.028	0.86	0.044	0.020	0.025	0.038	0.025	0.020
→ res (residual)	0.042	–	0.044	0.050	0.028 [†]	0.029 [†]	0.036	0.050
→ res-nplm (residual)	0.028	0.20	0.075	0.030	0.025	0.044	0.025	0.030
SupCon+classwise SIGReg ($\lambda=5$)	0.039	–	0.041	0.050	0.040	0.026 [†]	0.036	0.050
SupCon+SIGReg (repulse)	0.043	–	0.040	0.050	0.043	0.046	0.046	0.050
→ res (residual)	0.033	–	0.044	0.020	0.040	0.042	0.042	0.025
→ res-nplm (residual)	0.024	0.54	0.025	0.016	0.015	0.040	0.017	0.027

Table 21: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 9, density-ratio pool, derived cut; 16 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.036	0.07	> 0.1	0.023	> 0.1	0.018	0.028	0.024
→ res (residual)	0.030	–	0.063 [†]	–	> 0.1	0.042 [†]	0.030 [†]	–
→ res-nplm (residual)	0.048	–	> 0.1	–	> 0.1	0.028 [†]	0.052 [†]	–
SupCon+SIGReg ($\lambda=5$)	0.026	–	0.059 [†]	–	0.056 [†]	0.016 [†]	0.026 [†]	–
→ res (residual)	0.032	–	0.052 [†]	–	> 0.1	0.047 [†]	0.037 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.024 [†]	0.026 [†]	–
NPLM-dist. (sup.)	0.041	–	> 0.1	–	> 0.1	0.019 [†]	0.048 [†]	–
→ res (residual)	0.033	–	> 0.1	–	> 0.1	0.047 [†]	0.034 [†]	–
→ res-nplm (residual)	0.031	–	> 0.1	–	> 0.1	0.047 [†]	0.033 [†]	–
NPLM-dist.+classwise SIGReg	0.029	0.04	> 0.1	0.034	> 0.1	0.037	0.033	0.025
→ res (residual)	0.030	–	> 0.1	–	> 0.1	0.036 [†]	0.037 [†]	–
→ res-nplm (residual)	0.030	–	> 0.1	–	> 0.1	0.041 [†]	0.030 [†]	–

Table 22: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

CIFAR-100: discovery mode (np pool, derived cut), per draw

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.023	–	> 0.1	–	0.062 [†]	0.027 [†]	0.023 [†]	–
→ res (residual)	0.023	–	0.028 [†]	–	0.047 [†]	0.033 [†]	0.023 [†]	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	0.025 [†]	0.028 [†]	–
SupCon+SIGReg ($\lambda=5$)	0.022	–	0.082 [†]	–	0.059 [†]	0.027 [†]	0.021 [†]	–
→ res (residual)	0.026	–	0.027 [†]	–	0.040 [†]	0.029 [†]	0.026 [†]	–
→ res-nplm (residual)	0.024	–	> 0.1	–	> 0.1	0.028 [†]	0.023 [†]	–
NPLM-dist. (sup.)	0.046	–	> 0.1	–	> 0.1	0.046 [†]	0.062 [†]	–
→ res (residual)	0.026	–	0.019 [†]	–	0.038 [†]	0.025 [†]	0.026 [†]	–
→ res-nplm (residual)	0.026	–	0.084 [†]	–	> 0.1	0.033 [†]	0.027 [†]	–
NPLM-dist.+classwise SIGReg	0.027	–	> 0.1	–	> 0.1	0.033 [†]	0.029 [†]	–
→ res (residual)	0.024	–	0.027 [†]	–	> 0.1	0.025 [†]	0.025 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.031 [†]	0.028 [†]	–

Table 23: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 43, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.010	–	0.045 [†]	–	> 0.1	0.026 [†]	0.013 [†]	–
→ res (residual)	0.028	–	> 0.1	–	> 0.1	0.016 [†]	0.029 [†]	–
→ res-nplm (residual)	0.016	0.02	0.015	0.022	0.035	0.013	0.008	0.016
SupCon+SIGReg ($\lambda=5$)	0.014	0.12	0.024	0.011	0.044	0.017	0.016	0.016
→ res (residual)	0.028	–	> 0.1	–	> 0.1	0.027 [†]	0.030 [†]	–
→ res-nplm (residual)	0.016	0.08	0.025	0.014	0.038	0.016	0.014	0.015
NPLM-dist. (sup.)	0.017	0.35	0.033	0.024	0.029	0.016	0.026	0.010
→ res (residual)	0.025	–	0.036 [†]	–	0.031 [†]	0.017 [†]	0.023 [†]	–
→ res-nplm (residual)	0.021	–	0.048 [†]	–	0.025 [†]	0.015 [†]	0.023 [†]	–
NPLM-dist.+classwise SIGReg	0.015	0.40	0.026	0.014	0.025	0.016	0.015	0.008
→ res (residual)	0.025	–	0.077 [†]	–	0.039 [†]	0.018 [†]	0.025 [†]	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	0.023 [†]	0.026 [†]	–

Table 24: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 48, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.024	–	> 0.1	–	0.039 [†]	0.036 [†]	0.027[†]	–
→ res (residual)	0.029	–	> 0.1	–	> 0.1	0.017[†]	0.028 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.028 [†]	0.026[†]	–
SupCon+SIGReg ($\lambda=5$)	0.024	–	> 0.1	–	0.059 [†]	0.021 [†]	0.019[†]	–
→ res (residual)	0.025	–	> 0.1	–	> 0.1	0.025[†]	0.028 [†]	–
→ res-nplm (residual)	0.024	–	0.052 [†]	–	0.033 [†]	0.017[†]	0.021 [†]	–
NPLM-dist. (sup.)	0.026	0.07	0.037 [†]	0.023	0.037 [†]	0.025	0.018	0.022
→ res (residual)	0.026	–	> 0.1	–	> 0.1	0.022[†]	0.027 [†]	–
→ res-nplm (residual)	0.025	–	0.076 [†]	–	0.052 [†]	0.024[†]	0.026 [†]	–
NPLM-dist.+classwise SIGReg	0.023	0.06	0.034	0.035	0.026	0.016	0.016	0.010
→ res (residual)	0.029	–	> 0.1	–	> 0.1	0.023[†]	0.034 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.022[†]	0.026 [†]	–

Table 25: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 57, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.892	0.05	.029	.052	.078	.037	>.1	.015	.048	.031	.022	.023	.015	.073	.066	.035	.042
SupCon+SIGReg ($\lambda=5$)	0.762	0.13	.023	>.1	>.1	.033	>.1	.015	.029	.029	.040	.037	.014	>.1	>.1	.028	.043
NPLM-dist. (sup.)	0.787	0.13	>.1	>.1	>.1	.021	>.1	.019	.063	.037	.026	.024	.029	.032	.087	.022	.031
SimCLR	0.829	0.05	.029	.046	>.1	.036	.070	.043	.039	.070	.045	.047	.078	.081	>.1	.095	>.1
SimCLR+SIGReg	0.836	0.06	.048	.065	>.1	.034	>.1	.035	.029	.039	.080	.062	.046	.095	>.1	.046	.093
NPLM-bilinear	0.808	0.08	.078	.042	>.1	.026	>.1	.038	.039	.056	.046	.052	.052	.070	.093	.049	.088
GCD (transductive)	0.899	0.30	.009	.013	.009	.031	.017	.019	.025	.029	.024	.019	.009	.017	.021	.031	.025
GCD+SIGReg (transd.)	0.872	0.48	.015	.019	.019	.043	.025	.014	.035	.040	.042	.029	.013	.027	.026	.030	.028
SupCon $\rightarrow$ res (residual)	0.910	0.11	.085	.048	>.1	.024	.025	.021	.040	.045	.043	.039	.036	.063	.072	.044	.040
SupCon $\rightarrow$ res (concat)	0.945	0.12	.084	.041	>.1	.024	.034	.019	.040	.042	.042	.038	.029	.056	.046	.042	.026
SupCon $\rightarrow$ res-nplm (residual)	0.887	0.08	>.1	.027	>.1	.033	>.1	.016	.041	.073	.045	.032	.029	.072	.089	.040	.078
SupCon $\rightarrow$ res-nplm (concat)	0.931	0.09	.070	.034	.070	.034	.052	.015	.042	.031	.026	.024	.023	.063	.073	.031	.041
SupCon+SIGReg $\rightarrow$ res (residual)	0.854	0.06	>.1	.043	>.1	.019	.041	.019	.036	.045	.040	.031	.043	.089	.081	.037	.072
SupCon+SIGReg $\rightarrow$ res (concat)	0.855	0.09	.031	.063	>.1	.021	.042	.017	.026	.037	.036	.031	.019	.092	.085	.028	.041
SupCon+classwise SIGReg ($\lambda=5$)	0.810	0.05	>.1	.043	>.1	.068	.070	.027	.036	.039	.031	.052	.066	.052	>.1	.047	.052

Table 26: Galaxy10 / lejepa: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / LeJEPA, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

Galaxy10: every point, per backbone

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.019	.044	.038	.023	.027	.027
Eucl. (anch.)	.010	.020	.018	.014	.020	.018
Maha.	.021	.026	.045	.015	.070	.026
MMD	.010	.071	.046	.019	.040	.040
SparKer	.010	.047	.044	.028	.026	.028
SparKer (anch.)	.010	.036	.042	.025	.025	.025
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.012	.013	.023	.010	.020	.013
Maha.	.033	.034	.059	.015	.025	.033
MMD	.026	.052	.043	.049	.059	.049
SparKer	.027	.066	.048	.043	.045	.045
SparKer (anch.)	.015	.074	.040	.044	.028	.040
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.064	.033	.097	.017	>.1	.064
Eucl. (anch.)	.010	.020	.036	.013	.020	.020
Maha.	.031	.027	>.1	.019	.096	.031
MMD	.021	.045	.069	.047	.037	.045
SparKer	.026	.042	.069	.036	.052	.042
SparKer (anch.)	.017	.027	.042	.041	.028	.028
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.038	.070	.079	.024	.069	.069
Eucl. (anch.)	.010	.020	.016	.014	.020	.016
Maha.	.018	.047	.072	.023	.056	.047
MMD	.018	.047	.042	.028	.041	.041
SparKer	.017	.062	.084	.019	.043	.043
SparKer (anch.)	.010	.052	.069	.018	.028	.028
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.052	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.018	.025	.011	.020	.018
Maha.	.097	.039	>.1	.014	.029	.039
MMD	.026	.043	.041	.046	.045	.043
SparKer	.026	.088	.085	.041	.049	.049
SparKer (anch.)	.016	.056	.046	.029	.029	.029
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: never, d3: never, d5: 0.05, d7: 0.1, d8: 0.02; median engaged purity 0.59						
Eucl.	>.1	.070 [†]	>.1	.021 [†]	.029 [†]	.070
Eucl. (anch.)	—	—	—	—	.020	.020
Maha.	>.1	>.1	>.1	.021 [†]	.045	.150
MMD	.016 [†]	.052 [†]	.037	.027 [†]	.027 [†]	.027
SparKer	.029 [†]	.028 [†]	.088 [†]	.023 [†]	.028 [†]	.028
SparKer (anch.)	—	—	—	—	.020	.020

Table 27: Galaxy10 / lejepa: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / lejepa, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.906	0.10	>.1	.065	>.1	.042	>.1	.019	.031	.034	.029	.019	.031	.047	.065	.021	.030
SupCon+SIGReg ($\lambda=5$)	0.823	0.09	>.1	.065	>.1	.029	>.1	.025	.018	.033	.037	.033	.036	.046	.076	.027	.043
NPLM-dist. (sup.)	0.738	0.13	>.1	.065	>.1	.019	.052	.024	.065	.043	.028	.029	.032	.040	.094	.017	.024
SimCLR	0.847	0.04	.075	.047	>.1	.031	>.1	.049	.044	.066	.052	.049	.072	.093	.090	.082	>.1
SimCLR+SIGReg	0.835	0.05	>.1	.090	>.1	.038	>.1	.040	.026	.044	.070	.046	.069	.091	>.1	.042	.092
NPLM-bilinear	0.672	0.06	>.1	.096	>.1	>.1	>.1	.044	.033	.034	.059	.034	.069	.063	.047	.097	>.1
GCD (transductive)	0.886	0.40	.015	.009	.012	.041	.018	.024	.021	.026	.029	.028	.010	.015	.017	.027	.019
GCD+SIGReg (transd.)	0.852	0.48	.013	.017	.026	.031	.026	.017	.024	.039	.033	.027	.022	.022	.032	.029	.019
SupCon $\rightarrow$ res (residual)	0.930	0.06	>.1	.037	>.1	.043	.037	.016	.025	.035	.036	.036	.026	.087	.082	.047	.043
SupCon $\rightarrow$ res (concat)	0.947	0.09	>.1	.040	.084	.031	.042	.019	.026	.034	.026	.025	.025	.045	.063	.028	.029
SupCon $\rightarrow$ res-nplm (residual)	0.890	0.11	>.1	.052	>.1	.024	.073	.019	.036	.033	.047	.027	.031	.065	.081	.022	.029
SupCon $\rightarrow$ res-nplm (concat)	0.923	0.12	>.1	.048	>.1	.033	.075	.018	.035	.033	.029	.024	.029	.049	.071	.023	.029
SupCon+SIGReg $\rightarrow$ res (residual)	0.867	0.05	>.1	.056	>.1	.017	.052	.025	.024	.039	.045	.038	.038	.086	.080	.043	.062
SupCon+SIGReg $\rightarrow$ res (concat)	0.888	0.09	>.1	.042	>.1	.019	.036	.021	.021	.033	.040	.029	.029	.043	.052	.030	.037
SupCon+classwise SIGReg ($\lambda=5$)	0.898	0.07	>.1	.048	>.1	.042	.070	.023	.031	.052	.042	.028	.041	.069	.089	.047	.059

Table 28: Galaxy10 / visreg: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / VISREG, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.070	.038	.063	.024	.028	.038
Eucl. (anch.)	.015	.020	.021	.019	.020	.020
Maha.	>.1	.024	.056	.021	.042	.042
MMD	.017	.037	.040	.029	.027	.029
SparKer	.025	.056	.063	.024	.033	.033
SparKer (anch.)	.022	.042	.044	.021	.021	.022
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.020	.024	.014	.014	.015
Maha.	.045	.042	.066	.019	.033	.042
MMD	.031	.026	.029	.026	.045	.029
SparKer	.025	.054	.084	.038	.039	.039
SparKer (anch.)	.013	.041	.077	.024	.029	.029
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	.034	.037	.052	.022	.025	.034
Eucl. (anch.)	.015	.021	.025	.013	.019	.019
Maha.	.056	.024	.049	.015	.028	.028
MMD	.024	.040	.039	.042	.038	.039
SparKer	.026	.048	.047	.024	.024	.026
SparKer (anch.)	.017	.025	.044	.020	.027	.025
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.077	.048	>.1	.016	>.1	.077
Eucl. (anch.)	.017	.022	.016	.013	.020	.017
Maha.	>.1	.026	.097	.023	.059	.059
MMD	.019	.036	.039	.041	.034	.036
SparKer	.034	.030	.062	.026	.045	.034
SparKer (anch.)	.018	.029	.040	.017	.025	.025
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.02, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.018	.031	.010	.020	.020
Maha.	.063	.034	.072	.015	.027	.034
MMD	.028	.033	.043	.038	.029	.033
SparKer	.031	.052	.078	.044	.047	.047
SparKer (anch.)	.021	.034	.042	.037	.026	.034
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: 0.02, d3: 0.05, d5: 0.1, d7: 0.03, d8: 0.1; median engaged purity 0.43						
Eucl.	.082 [†]	.041	>.1	.015 [†]	.026 [†]	.041
Eucl. (anch.)	—	—	—	.030	—	.030
Maha.	>.1	.039	>.1	.016 [†]	.044 [†]	.044
MMD	.018	.044	.044 [†]	.025	.037 [†]	.037
SparKer	.034 [†]	.032	.087	.017 [†]	.027 [†]	.032
SparKer (anch.)	—	—	—	.030	—	.030

Table 29: Galaxy10 / visreg: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / visreg, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.879	0.06	>.1	>.1	.063	.076	>.1	.031	.029	.025	.042	.037	.078	>.1	>.1	.049	.083
SupCon+SIGReg ($\lambda=5$)	0.804	0.04	>.1	.072	.069	>.1	>.1	.021	.025	.031	.047	.029	>.1	>.1	>.1	.079	>.1
NPLM-dist. (sup.)	0.775	0.10	>.1	>.1	>.1	.059	>.1	.037	.023	.029	.038	.037	.054	.086	.070	.038	.048
SimCLR	0.790	0.05	>.1	.066	>.1	>.1	>.1	.029	.039	.056	.067	.052	.070	.097	>.1	.066	>.1
SimCLR+SIGReg	0.801	0.04	>.1	.069	>.1	.076	>.1	.027	.029	.041	.088	.059	.069	.084	>.1	.059	.091
NPLM-bilinear	0.684	0.04	>.1	>.1	.082	>.1	>.1	.097	.042	.041	.093	>.1	.081	.089	.079	.083	.089
GCD (transductive)	0.821	0.07	>.1	.052	.037	.076	>.1	.019	.025	.031	.040	.028	.059	.087	.085	.069	.092
GCD+SIGReg (transd.)	0.831	0.12	.084	.066	.056	.073	>.1	.024	.026	.034	.039	.034	.052	>.1	.096	.066	.098
SupCon $\rightarrow$ res (residual)	0.891	0.05	>.1	.080	.062	.097	>.1	.023	.029	.047	.056	.033	.090	.098	.097	.081	.097
SupCon $\rightarrow$ res (concat)	0.903	0.06	>.1	>.1	.063	.074	>.1	.025	.025	.029	.048	.034	.074	>.1	.098	.073	.092
SupCon $\rightarrow$ res-nplm (residual)	0.862	0.07	>.1	>.1	.056	.087	>.1	.024	.024	.036	.039	.032	.077	>.1	>.1	.047	.092
SupCon $\rightarrow$ res-nplm (concat)	0.890	0.06	>.1	>.1	.063	.075	>.1	.029	.026	.027	.042	.033	.048	.093	>.1	.047	.092
SupCon+SIGReg $\rightarrow$ res (residual)	0.852	0.06	>.1	.073	.092	>.1	>.1	.025	.025	.044	.049	.032	.088	>.1	.096	.082	>.1
SupCon+SIGReg $\rightarrow$ res (concat)	0.869	0.05	>.1	.079	.072	.096	>.1	.021	.025	.036	.047	.033	.089	>.1	>.1	.075	>.1
SupCon+classwise SIGReg ($\lambda=5$)	0.832	0.05	>.1	.059	.077	>.1	>.1	.021	.023	.039	.049	.077	.096	.096	.087	.070	.073

Table 30: Galaxy10 / dino: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / DINO, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.03, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.090	.022	>.1	>.1	.150
Eucl. (anch.)	.019	.021	.030	.014	.018	.019
Maha.	>.1	>.1	.066	>.1	>.1	.150
MMD	.029	.039	.036	.059	.044	.039
SparKer	.062	.079	>.1	.048	.079	.079
SparKer (anch.)	.044	.044	.089	.035	.042	.044
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: 0.02, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.078	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.021	.025	.020	.031	.021
Maha.	>.1	.070	.069	.085	>.1	.085
MMD	.052	.028 [†]	.066	.087	.066	.066
SparKer	.040	.083	.096	.078	>.1	.083
SparKer (anch.)	.039	.036	.049	.044	.044	.044
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.02, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.072	.036	>.1	>.1	.150
Eucl. (anch.)	.016	.019	.038	.020	.019	.019
Maha.	>.1	>.1	.041	.068	>.1	.150
MMD	.027	.039	.040	.050	.036	.039
SparKer	.046	.084	.092	.069	.097	.084
SparKer (anch.)	.045	.037	.052	.044	.043	.044
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: never, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.062	>.1	>.1	.150
Eucl. (anch.)	.020	.019	.022	—	.031	.021
Maha.	>.1	.075	.078	>.1	>.1	.150
MMD	.027	.029	.039	.046 [†]	.046	.039
SparKer	.048	.077	.084	.075 [†]	>.1	.077
SparKer (anch.)	.044	.046	.047	—	.052	.047
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.096 [†]	>.1	>.1	.150
Eucl. (anch.)	.014	.026	.026	—	.034	.026
Maha.	.097	.086	>.1	>.1	>.1	.150
MMD	.066	.028 [†]	.047	.052 [†]	.048	.048
SparKer	.059	>.1	.074 [†]	.084 [†]	>.1	.084
SparKer (anch.)	.047	.074	.043	—	.046	.047
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: never, d3: never, d5: never, d7: never, d8: 0.03; median engaged purity 0.28						
Eucl.	.071 [†]	>.1	>.1	.018 [†]	>.1	.150
Eucl. (anch.)	—	—	—	—	.030	.030
Maha.	>.1	>.1	>.1	.072 [†]	>.1	.150
MMD	.035 [†]	.024 [†]	.029 [†]	.037 [†]	.052	.035
SparKer	.045 [†]	.078 [†]	.056 [†]	.045 [†]	.094	.056
SparKer (anch.)	—	—	—	—	.047	.047

Table 31: Galaxy10 / dino: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / dino, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 168 (test, draw) points.

space	one label (class 2)					all seen labels				
	out. AUC	probe	Maha.	MMD	SparKer	out. AUC	probe	Maha.	MMD	SparKer
<i>lejepa</i>										
pretrained trunk	0.86	1.00	.007	.042	.022	0.89	1.00	.006	.023	.042
SupCon	0.69	0.99	.079	>.1	.036	0.51	0.95	.016	.083	>.1
SupCon+SIGReg ($\lambda=5$)	0.57	0.99	.028	>.1	.035	0.57	0.94	.031	>.1	.082
NPLM-dist. (sup.)	0.48	0.88	.013	>.1	>.1	0.58	0.86	.027	>.1	>.1
<i>visreg</i>										
pretrained trunk	0.94	1.00	.006	.047	.008	0.94	1.00	.006	.024	.016
SupCon	0.52	1.00	.022	>.1	.023	0.59	1.00	.022	.083	.056
SupCon+SIGReg ($\lambda=5$)	0.84	1.00	.007	>.1	.009	0.69	1.00	.013	>.1	.037
NPLM-dist. (sup.)	0.35	0.77	.093	>.1	>.1	0.62	0.64	.068	.086	.088
<i>dino</i>										
pretrained trunk	0.77	1.00	.010	.049	.020	0.82	1.00	.008	.027	.080
SupCon	0.61	1.00	.013	>.1	.023	0.63	1.00	.010	.076	.060
SupCon+SIGReg ($\lambda=5$)	0.68	1.00	.024	>.1	.024	0.63	0.99	.022	.059	.067
NPLM-dist. (sup.)	0.76	0.81	.031	>.1	.043	0.63	0.69	.063	>.1	.086

Table 32: Diffuse-shift detection: a systematic smear of seen-class images instead of a novel class. The smear stays almost perfectly *decodable* (probe 0.86–1.00) but the fine-tuned discovery spaces barely see it as outlying; mean-shift Mahalanobis becomes the only reliable test there, while the raw pretrained trunk detects it at $f^*=0.006$ –0.010 everywhere — the systematics monitor. SparKer prefers the semi-clustered one-label smear in every engaged comparison: the kernel tests’ class-novelty sharpness is a dependence on clusteredness. *Coverage*: Galaxy10, 24 (backbone, space, mode) cells; smear = Gaussian blur $\sigma=2$ px applied identically to ≤ 1000 seen-class test images (source rows removed from the background pool); exp-146 toy battery. *out. AUC* = min-anchor-distance ROC of smeared vs. background; *probe* = smeared-vs-clean linear AUC.

Galaxy10: diffuse-shift boundary

space	novel class 2 (held out)			smeared class 2 (seen, blur $\sigma=2$)			cost
	Maha.	MMD	SparKer	Maha.	MMD	SparKer	
<i>lejepa</i>							
pretrained trunk	>.1	.031	.079	.007	.042	.022	0.2×
SupCon	.029	.015	.015	.079	>.1	.036	2.4×
SupCon+SIGReg ($\lambda=5$)	.023	.015	.014	.028	>.1	.035	1.9×
NPLM-dist. (sup.)	>.1	.019	.029	.013	>.1	>.1	0.7×
<i>visreg</i>							
pretrained trunk	>.1	.034	.073	.006	.047	.008	0.2×
SupCon	>.1	.019	.031	.022	>.1	.023	1.1×
SupCon+SIGReg ($\lambda=5$)	>.1	.025	.036	.007	>.1	.009	0.3×
NPLM-dist. (sup.)	>.1	.024	.032	.093	>.1	>.1	3.9×
<i>dino</i>							
pretrained trunk	>.1	.040	.049	.010	.049	.020	0.3×
SupCon	>.1	.031	.078	.013	>.1	.023	0.4×
SupCon+SIGReg ($\lambda=5$)	>.1	.021	>.1	.024	>.1	.024	1.1×
NPLM-dist. (sup.)	>.1	.037	.054	.031	>.1	.043	0.8×

Table 33: Same label, two anomaly types: class novelty vs. diffuse smear. The reduction is per test family, not blanket: the clustering-sensitive tests lose heavily on the smear (MMD censors on 8 of 9 fine-tuned rows; SparKer costs 1.5–2.4× where both engage), while the mean-shift Mahalanobis often detects the smear *cheaper* than novelty (costs down to 0.3×) — but a mean-shift detection carries no class structure for an anchor to mean. The raw pretrained trunk sees the smear 3–5× better than it sees the same class as novelty (cost 0.2–0.3×): the systematics monitor. *Coverage:* Galaxy10, 12 (backbone, space) pairs, both anomalies built from CLASS 2: left = class 2 held out of training and injected as a novel class (exp-150 draw d0); right = class 2 seen in training, a subset smeared and injected (exp 152). The pretrained-trunk row uses the identical space on both sides; the fine-tuned rows differ in which class the fine-tune excluded (2 vs. the archived 9). cost = ratio of best-test f^* , smeared over novel.

backbone	probe	eucl	per-ev.	Maha.					MMD					SparKer				
				d0	d1	d2	d3	d4	d0	d1	d2	d3	d4	d0	d1	d2	d3	d4
<i>cub</i> — held-out classes d0=c133, d1=c89, d2=c53, d3=c136, d4=c146																		
lejepa	0.945	0.581	0.12	.023	>.1	>.1	.014	.029	.041	.017	.017	.087	.040	.017	.008	.016	.020	.015
visreg	0.936	0.550	0.11	.026	>.1	>.1	.014	.022	.039	.017	.017	.087	.037	.018	.008	.018	.017	.016
dino	0.988	0.632	0.11	.072	>.1	.023	.009	.034	.015	.017	.009	.029	.018	.010	.006	.010	.016	.012
<i>aircraft</i> — held-out classes d0=c14, d1=c43, d2=c71, d3=c96, d4=c58																		
lejepa	0.917	0.580	0.08	>.1	.023	>.1	>.1	.036	.097	.068	.047	.075	.056	.065	.036	.033	.035	.039
visreg	0.926	0.561	0.08	>.1	.021	>.1	>.1	.040	>.1	.065	.047	.073	.056	.047	.031	.034	.037	.036
dino	0.947	0.649	0.12	>.1	.029	.069	.043	.019	.082	.056	.028	.042	.036	.041	.038	.019	.026	.023
<i>cars</i> — held-out classes d0=c172, d1=c13, d2=c109, d3=c112, d4=c53																		
lejepa	0.876	0.469	0.02	.075	>.1	>.1	>.1	>.1	.028	.034	.027	.026	.027	.015	.021	.019	.015	.015
visreg	0.932	0.472	0.02	.085	>.1	>.1	>.1	>.1	.027	.028	.033	.026	.028	.019	.019	.018	.014	.019
dino	0.971	0.472	0.03	.066	>.1	.079	>.1	>.1	.038	.035	.029	.021	.021	.014	.018	.013	.012	.013

Table 34: The on-manifold corner, every point: the Semantic Shift Benchmark fine-grained triplet (CUB-200 birds, FGVC-Aircraft variants, Stanford car models), novelty inside the seen manifold. A *spectrum*, not a single failure: on all three the probe is near-perfect (0.88–0.99) yet mean-shift Mahalanobis censors and per-event power is dead — but locality survives, and how well tracks how separable the novel clump is. CUB is the mildest (eucl AUC 0.55–0.63, above chance; SparKer $f^*=0.010$ –0.017; even MMD engages), aircraft is intermediate (SparKer 0.026–0.036), and cars is the hardest — min-anchor-distance ranks the novel model *below* chance (eucl AUC ~ 0.47) yet SparKer still detects it ($f^*=0.013$ –0.019). Detection survives throughout; nothing pool-able or anchor-able comes with it. *Coverage:* Pretrained ViT-B/16 trunk (identity head), single holdout, 5 draws per (dataset, backbone), 45 cells, every point shown; probe/eucl/per-ev. are means over draws. Held-out pools are tiny (aircraft ~ 33 test images/class), so toys bootstrap them — censoring here is conservative evidence.

arm	frozen f^*			eng. f	pur.	post-discovery $f^*(2\sigma)$					
	Maha.	MMD	SpK			Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
<i>aircraft</i> — last 10 classes held out (exp-70 split)											
<i>lejepa</i>											
SupCon	.036	.068	.059	0.05	0.86	0.073	0.050	0.039	0.052	0.076	0.073
SupCon $\rightarrow$ res	.077	.081	>.1	0.1	0.88	> 0.1	—	0.097	0.079	0.089	—
SupCon $\rightarrow$ res-nplm	>.1	.085	>.1	0.1	0.86	> 0.1	—	0.077	0.083	0.098	—
SupCon+SIGReg ($\lambda=5$)	.041	.073	.088	0.05	0.86	0.071	0.050	0.038	0.082	0.079	0.069
SupCon+SIGReg $\rightarrow$ res	>.1	>.1	>.1	0.1	0.67	> 0.1	—	0.098	0.087	0.086	—
<i>visreg</i>											
SupCon	.028	.052	.052	0.05	1.00	0.035	0.050	0.028 [†]	0.072	0.047	0.056
SupCon $\rightarrow$ res	.073	.056	.087	0.05	0.75	0.066	0.050	0.052	0.049	0.049	0.059
SupCon $\rightarrow$ res-nplm	.031	.044	.029	0.01	0.58	0.019	0.017	0.026	0.040	0.038	0.076
SupCon+SIGReg ($\lambda=5$)	.039	.044	.079	0.1	1.00	0.080	—	0.038 [†]	0.037 [†]	0.073	—
SupCon+SIGReg $\rightarrow$ res	>.1	.088	>.1	0.1	0.93	> 0.1	—	0.095	0.083	> 0.1	—
<i>dino</i>											
SupCon	.063	.072	.092	0.05	0.86	0.043	0.052	0.074	0.056	0.084	0.080
SupCon $\rightarrow$ res	.066	.081	.085	0.05	0.67	0.047	0.052	0.077	0.059	0.067	0.059
SupCon $\rightarrow$ res-nplm	>.1	.045	.090	0.05	0.71	0.037	0.050	0.067	0.056	0.070	0.050
SupCon+SIGReg ($\lambda=5$)	.062	.056	.085	0.05	0.86	0.093	0.050	0.064	0.059	0.069	0.062
SupCon+SIGReg $\rightarrow$ res	>.1	.096	>.1	0.1	0.54	> 0.1	—	> 0.1	0.081	0.098	—
<i>cars</i> — last 10 classes held out (exp-70 split)											
<i>lejepa</i>											
SupCon	.033	.049	.046	never	—	0.039 [†]	—	0.025 [†]	0.048 [†]	0.044 [†]	—
SupCon $\rightarrow$ res	>.1	.075	.083	never	—	> 0.1	—	> 0.1	0.082 [†]	0.091 [†]	—
SupCon $\rightarrow$ res-nplm	>.1	.056	.092	never	—	> 0.1	—	> 0.1	0.047 [†]	0.098 [†]	—
SupCon+SIGReg ($\lambda=5$)	.063	.049	.056	never	—	0.075 [†]	—	0.052 [†]	0.048 [†]	0.069 [†]	—
SupCon+SIGReg $\rightarrow$ res	>.1	.049	.083	never	—	> 0.1	—	> 0.1	0.073 [†]	0.084 [†]	—
<i>visreg</i>											
SupCon	.025	.048	.037	0.1	1.00	0.021 [†]	—	0.023 [†]	0.045 [†]	0.029 [†]	—
SupCon $\rightarrow$ res	.044	.034	.028	0.1	0.67	0.065	—	0.042 [†]	0.047 [†]	0.026 [†]	—
SupCon $\rightarrow$ res-nplm	.017	.040	.019	0.02	0.36	0.022	0.020	0.019	0.032	0.023	0.048
SupCon+SIGReg ($\lambda=5$)	.045	.049	.052	never	—	0.077 [†]	—	0.052 [†]	0.048 [†]	0.059 [†]	—
SupCon+SIGReg $\rightarrow$ res	>.1	.052	.093	never	—	> 0.1	—	> 0.1	0.052 [†]	0.087 [†]	—
<i>dino</i>											
SupCon	.033	.056	.040	never	—	0.035 [†]	—	0.031 [†]	0.048 [†]	0.044 [†]	—
SupCon $\rightarrow$ res	.036	.047	.028	0.05	0.64	0.043	0.050	0.034	0.049	0.033	0.050
SupCon $\rightarrow$ res-nplm	.023	.052	.025	0.02	0.07	0.043	0.037	0.031	0.047	0.025 [†]	0.037
SupCon+SIGReg ($\lambda=5$)	.043	.063	.040	never	—	0.056 [†]	—	0.048 [†]	0.062 [†]	0.042 [†]	—
SupCon+SIGReg $\rightarrow$ res	>.1	.097	>.1	never	—	> 0.1	—	> 0.1	> 0.1	> 0.1	—

Table 35: Discovery mode on the on-manifold corner. The two datasets split: on *aircraft* the derived cut declines at $f \leq 0.03$ and engages at $f=0.05$ with purity 0.54–1.00 on every backbone (visreg res-nplm engages at 0.01 and posts eucl-disc 0.017); on *cars* the parents’ critics never engage on 2 of 3 backbones and only the res-nplm child pools anything — and even where anchors appear, the post numbers do not beat the frozen SparKer, so discovery adds nothing beyond locality there. Residual children are what softens the corner. *Coverage:* Exp 154; 30 (dataset, backbone, arm) cells, fine-tuned exp-70 heads (10 of 100/196 classes held out — the multi-holdout regime; never pool with the single-holdout Table 34), head-only loop, np pool + derived cut ($n_{\min}=5$), skip-on-refuse; injections clamp at the held-out train pool (cars: 432 rows, so $f=0.1$ realises ~ 0.053). [†] = cut declined at the crossing fraction (the frozen space’s number).

arm	eng. f	pur.	out. AUC (post)	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
<i>lejepa</i> , one label (class 2)									
SupCon	0.01	0.30	0.51–0.90	0.017	0.014	0.014	0.089	0.017	0.028
SupCon $\rightarrow$ res	0.02	0.70	0.39–0.48	> 0.1	0.020	0.038	> 0.1	0.045	0.046
SupCon $\rightarrow$ res-nplm	0.02	1.00	0.69–0.96	0.016	0.020	0.012	0.052	0.016	0.020
SupCon+SIGReg ($\lambda=5$)	0.01	0.64	0.76–0.83	0.022	0.010	0.014	> 0.1	0.016	0.013
SupCon+SIGReg $\rightarrow$ res	0.02	0.60	0.34–0.54	> 0.1	0.020	0.046	> 0.1	0.044	0.045
<i>lejepa</i> , all seen labels									
SupCon	0.02	0.19	0.51–0.79	0.044	0.035	0.021 [†]	0.081	0.045	0.040
SupCon $\rightarrow$ res	0.01	0.70	0.43–0.50	> 0.1	0.035	0.046	> 0.1	> 0.1	0.038
SupCon $\rightarrow$ res-nplm	0.01	0.86	0.86–0.94	0.012	0.021	0.010	0.029	0.018	0.019
SupCon+SIGReg ($\lambda=5$)	0.02	1.00	0.57–0.66	0.048	0.020	0.024	> 0.1	0.049	0.040
SupCon+SIGReg $\rightarrow$ res	0.02	0.47	0.34–0.42	> 0.1	0.079	0.073	> 0.1	0.085	0.087
<i>visreg</i> , one label (class 2)									
SupCon	0.01	1.00	0.66–0.88	0.024	0.016	0.010	> 0.1	0.014	0.018
SupCon $\rightarrow$ res	0.01	1.00	0.42–0.62	> 0.1	0.016	0.015	> 0.1	0.017	0.017
SupCon $\rightarrow$ res-nplm	0.01	1.00	0.79–0.93	0.016	0.010	0.010	0.090	0.010	0.017
SupCon+SIGReg ($\lambda=5$)	0.01	1.00	0.89–0.96	0.010	0.010	0.010	0.084	0.010	0.010
SupCon+SIGReg $\rightarrow$ res	0.02	1.00	0.49–0.64	> 0.1	0.020	0.016	0.089	0.019	0.020
<i>visreg</i> , all seen labels									
SupCon	0.01	0.57	0.72–0.82	0.024	0.018	0.012	0.059	0.026	0.028
SupCon $\rightarrow$ res	0.01	0.07	0.45–0.56	> 0.1	0.026	0.033	0.037	0.037	0.028
SupCon $\rightarrow$ res-nplm	0.01	0.15	0.66–0.81	0.024	0.028	0.016	0.086	0.027	0.047
SupCon+SIGReg ($\lambda=5$)	0.01	1.00	0.79–0.86	0.019	0.015	0.010	0.056	0.026	0.027
SupCon+SIGReg $\rightarrow$ res	0.05	1.00	0.35–0.47	> 0.1	0.050	0.041	0.056	0.045	0.050
<i>dino</i> , one label (class 2)									
SupCon	0.02	1.00	0.61–0.89	0.026	0.020	0.013	0.046	0.016	0.020
SupCon $\rightarrow$ res	0.02	1.00	0.52–0.80	0.025	0.020	0.015	0.056	0.023	0.020
SupCon $\rightarrow$ res-nplm	0.02	1.00	0.55–0.86	0.042	0.020	0.015	0.050	0.019	0.028
SupCon+SIGReg ($\lambda=5$)	0.02	1.00	0.68–0.75	0.041	0.020	0.016	0.079	0.018	0.020
SupCon+SIGReg $\rightarrow$ res	0.01	1.00	0.61–0.74	0.042	0.010	0.019	> 0.1	0.040	0.047
<i>dino</i> , all seen labels									
SupCon	0.01	0.39	0.82–0.93	0.024	0.015	0.010	0.025	0.028	0.026
SupCon $\rightarrow$ res	0.01	1.00	0.75–0.83	0.025	0.011	0.010	0.045	0.045	0.017
SupCon $\rightarrow$ res-nplm	0.01	1.00	0.65–0.89	0.059	0.054	0.010	0.087	0.095	0.059
SupCon+SIGReg ($\lambda=5$)	0.01	1.00	0.70–0.77	0.034	0.012	0.015	0.038	0.044	0.017
SupCon+SIGReg $\rightarrow$ res	0.01	1.00	0.60–0.69	0.039	0.011	0.021	0.046	0.085	0.047

Table 36: Discovery mode on the diffuse shift: the loop finds what the frozen space cannot see. Pre-discovery the smear is nearly invisible in these head spaces (Table 32); with the loop, the density-ratio pool isolates the smeared events in every cell (engagement at $f=0.01$ – 0.02 in 29 of 30, pool purity frequently 1.00 despite outlier AUC near chance), the fine-tune pulls the smear off the manifold (post outlier AUC up to 0.96), and the anchor-aware and mean tests reach $f^*\approx 0.01$ – 0.02 . MMD stays the blind test. The diffuse cell of the failure 2×2 is thus a *frozen-space* boundary, not a pipeline one — a mean-shift detection still carries no class semantics, but the anchors let the analyst inspect the isolated events. *Coverage*: Exp 155; 30 (backbone, mode, arm) cells. A fraction f of the seen train corpus is REPLACED by its blur- $\sigma=2$ smeared version (pseudo-labelled unlabelled-anomalous); the settled loop (np pool, derived cut, $n_{\min}=5$, skip-on-refuse) runs head-only; the signal pool is the exp-152 smeared test subset (same rng). pur. = smeared fraction of the round-1 pool; out. AUC = post-loop min-anchor-distance ROC of smeared vs. background.